Grade 9 – Mathematics

Third Term Model Paper 6

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Name / Index No.: ..... **Time: 3 hours**

- This paper consists of two parts: Part I and Part II.
- Part I: Answer **all** questions on this paper itself. Part A has 15 questions (2 marks each). Part B has 2 questions (10 marks each).
- Part II: Answer **five** questions only, on separate sheets. Each question carries 10 marks.
- Show all relevant steps of working. Write units where necessary. Use π = 22/7 where needed.

For examiner's use only	
PART I – A	/ 30
PART I – B	/ 20
PART II	/ 50
Total	**/ 100**

PART I

Part A – Answer all questions in the space provided. (2 marks each)

1. Write the general term of the number pattern 1, 4, 9, 16, … and find its 12th term. (2)

Answer: .....

2. Write the largest four-digit binary number and its value in base ten. (2)

Answer: .....

3. Simplify: 5/6 – 1/4 × 2/3 (2)

Answer: .....

4. The price of an item was reduced from Rs 800 to Rs 680. Find the percentage reduction. (2)

Answer: .....

5. Factorise: $a^2 - 2a - 15$ (2)

Answer: .....

6. Write 1/8 as a power of 2. (2)

Answer: .....

7. Write 6.05×10^{-3} as an ordinary decimal number. (2)

Answer: .....

8. Solve: $0.5x + 3 = 8$ (2)

Answer: .....

9. Given $2a = 14$, we get $a = 7$. Write the axiom used. (2)

Answer: .....

10. Three adjacent angles on a straight line are $2x$, $3x$ and $4x$. Find the value of x. (2)

Answer: .....

11. Find the perimeter of a quadrant (quarter of a circle) of radius 14 cm. ($\pi = 22/7$) (2)

Answer: .....

12. The length of a rectangle is 12 cm and its diagonal is 13 cm. Find its breadth. (2)

Answer: .....

13. Solve: $x/3 - 1 \geq 2$ (2)

Answer: .....

14. A = {a, b, c}. Write all the subsets of A that have exactly two elements. (2)

Answer: .....

15. Each interior angle of a regular polygon is 160°. Find the number of sides. (2)

Answer: .....

Part B – Answer both questions on this paper. (10 marks each)

1.

A and B are two posts 8 m apart on flat ground. A dog is tied to post A with a rope 5 m long. A water bowl P is to be placed where the dog can just reach it with the rope stretched, and at an equal distance from both posts.

(i) Name the locus of the points that are 5 m from A. (2 marks)

(ii) Name the locus of the points that are equidistant from A and B. (2 marks)

(iii) Draw a rough sketch showing both loci and mark the possible positions of P. (2 marks)

(iv) Using Pythagoras' relation, find the distance from P to the line AB. (2 marks)

(v) Find the area of the triangle APB. (2 marks)

2.

A cuboid-shaped container measuring 40 cm × 25 cm × 30 cm is full of fruit juice.

(i) Find the volume of the container in cm^3. (2 marks)

(ii) Find the amount of juice in litres. (2 marks)

(iii) How many 250 ml cups can be filled with this juice? (2 marks)

(iv) The juice costs Rs 120 per litre to make. Find the total cost. (2 marks)

(v) Each cup is sold for Rs 40. Find the profit as a percentage of the cost. (2 marks)

PART II

Answer five questions only. (10 marks each)

1.

(a) After an increase of 12%, a man's salary is Rs 56 000. Find his salary before the increase. (3 marks)

(b) A shopkeeper sells an article for Rs 1380 and makes a profit of 15%. Find the cost price. (3 marks)

(c) In a school, 40% of the students are boys and there are 360 girls. Find the total number of students and the number of boys. (4 marks)

2.

(a) Factorise: (i) $25 - x^2$ (ii) $3x^2 - 12x$ (iii) $x^2 + 10x + 25$ (5 marks)

(b) Simplify: $(2x + 1)/3 - (x - 2)/6$ (3 marks)

(c) If $x + y = 7$ and $x - y = 3$, find the value of $x^2 - y^2$ without finding x and y. (2 marks)

3.

The length y (cm) of a burning candle after x hours is shown in the table below.

Time x (hours)	0	1	2	3	4
Length y (cm)	20	17	14	11	8

(i) Draw the graph of y against x on graph paper. (3 marks)

(ii) Find the gradient of the graph and explain what it means. (2 marks)

(iii) Write the equation of the line in the form $y = mx + c$. (2 marks)

(iv) Find the length of the candle after 2.5 hours. (1 marks)

(v) After how many hours will the candle burn out completely? (2 marks)

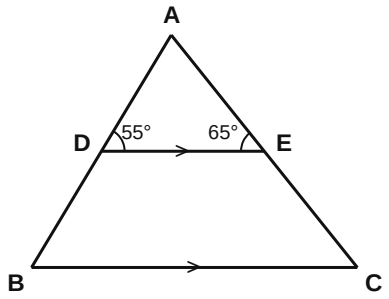
4.

(a) The angles of a triangle are $(x + 10)°$, $(2x - 20)°$ and $(x + 30)°$. Find x and the three angles. (3 marks)

(b) The price of 3 kg of sugar and 2 kg of dhal is Rs 1440. The price of 2 kg of sugar and 3 kg of dhal is Rs 1610. Find the price of 1 kg of each. (5 marks)

(c) Make b the subject of the formula $P = 2(l + b)$, and find b when $P = 50$ and $l = 15$. (2 marks)

5.



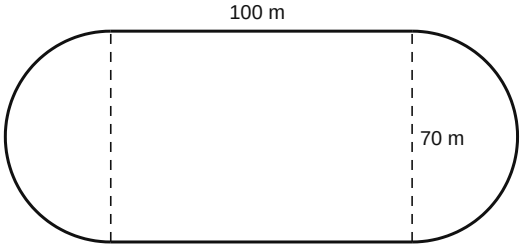
(a) In the figure, DE ∥ BC, AD̂E = 55° and AÊD = 65°. Find AB̂C, AĈB and BÂC, giving reasons. (4 marks)

(b) Find the sum of the interior angles of a hexagon, and the size of each interior angle of a regular hexagon. (2 marks)

(c) The exterior angles of a quadrilateral are x, 2x, 3x and 4x. Find x and the largest interior angle of the quadrilateral. (4 marks)

6.

(π = 22/7)



(a) The figure shows a running track made of two straight parts, each 100 m long, and two semicircular ends of diameter 70 m.

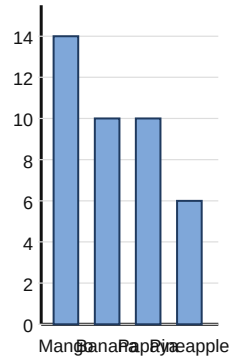
(i) Find the perimeter of the track. (3 marks)

(ii) Find the area enclosed by the track. (3 marks)

(b) The area of a parallelogram ABCD is 96 cm² and AB = 12 cm. (i) Find the perpendicular distance between AB and DC. (ii) If BC = 10 cm, find the perpendicular distance between BC and AD. (4 marks)

7.

The bar chart shows the favourite fruits of a group of students.



(i) How many students are in the group? (1 marks)

(ii) Write the modal fruit. (1 marks)

(iii) What percentage of the students like bananas? (2 marks)

(iv) If this information is shown in a pie chart, find the angle of the sector for mango. (3 marks)

(v) A student is chosen at random. Find the probability that the student's favourite fruit is papaya. (2 marks)

(vi) How many more students like mango than pineapple? (1 marks)